

Python Module 2 – Theory Questions & Answers

Q1. How memory is managed in Python?

Answer: Python manages memory automatically using a private memory manager. It stores objects in memory and removes unused objects using garbage collection. The programmer usually does not need to free memory manually.

Q2. What is the purpose of continue statement in Python?

Answer: The continue statement is used inside loops. It skips the current iteration and moves to the next iteration of the loop.

Q3. What are negative indexes and why are they used?

Answer: Negative indexes are used to access elements from the end of a string, list, or tuple. For example, -1 refers to the last element, -2 refers to the second last element. They are useful when we want to access data from the end.

Q4. What is List? How will you reverse a list?

Answer: A list is a collection in Python used to store multiple items. Lists are ordered and mutable. A list can be reversed using reverse() method or slicing like list[::-1].

Q5. How will you remove last object from a list?

Answer: The last object from a list can be removed using the pop() method without giving any index.

Q6. Suppose list1 is [2, 33, 222, 14, 25], what is list1[-1]?

Answer: list1[-1] returns the last element of the list, which is 25.

Q7. Differentiate between append() and extend() methods.

Answer: append() adds a single item at the end of the list, while extend() adds multiple items from another iterable to the list.

Q8. How will you compare two lists?

Answer: Two lists can be compared using comparison operators like ==. If all elements and their order are the same, the lists are equal.

Q9. What is tuple? Difference between list and tuple.

Answer: A tuple is a collection of items in Python that is ordered but immutable. Lists can be changed after creation, while tuples cannot be changed.

Q10. How will you create a dictionary using tuples in Python?

Answer: A dictionary can be created using tuples with the dict() function. Each tuple contains a key and value pair.

Q11. How Do You Traverse Through A Dictionary Object In Python?

Answer: A dictionary can be traversed using loops like for loop with methods such as items(), keys(), and values().

Q12. How Do You Check The Presence Of A Key In A Dictionary?

Answer: The presence of a key can be checked using the in keyword.

Q13. Why Do You Use the zip() Method in Python?

Answer: The zip() method is used to combine multiple iterables like lists or tuples element by element.

Q14. How do you perform pattern matching in Python? Explain.

Answer: Pattern matching in Python can be done using regular expressions with the re module. It helps to search and match patterns in strings.

Q15. What is lambda function in Python? What do we call an incomplete version of a function?

Answer: A lambda function is a small anonymous function written in a single line using the lambda keyword. An incomplete version of a function is often called a lambda or anonymous function.

Q16. How Many Basic Types Of Functions Are Available In Python?

Answer: There are mainly two basic types of functions in Python: built-in functions and user-defined functions.

Q17. How can you pick a random item from a list or tuple?

Answer: A random item can be selected using random.choice() function.

Q18. How can you pick a random item from a range?

Answer: A random item from a range can be selected using random.randrange() or random.choice().

Q19. How can you get a random number in Python?

Answer: Random numbers can be generated using the random module, such as `random.randint()`.

Q20. How will you set the starting value in generating random numbers?

Answer: The starting value can be set using `random.seed()`. It helps generate the same random sequence every time.

Q21. How will you randomize the items of a list in place?

Answer: The items of a list can be randomized using `random.shuffle()`.